LF3 Worked Examples: Graphing Lines

Recording companion for the LF3 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What two points can I plot? Every line needs two points. Intercepts give points with a 0 in them: an x-intercept of 3 is the point (3, 0). A slope gives a second point from the first: a slope of −2/5 means right 5, down 2. From an equation, find a point and the slope, or both intercepts. x = c is a vertical line and y = c is a horizontal one.

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 1: Two Intercepts

x-intercept −3
y-intercept 5

Question

Which two points do you plot to graph this line?

A. (0, −3) and (5, 0)
B. (−3, 0) and (0, 5)
C. (−3, 5) only
D. (3, 0) and (0, 5)

Answer and reasoning

An intercept is where the line crosses an axis. The *x*-intercept is on the *x*-axis, where *y=0*: *(−3,0)*. The *y*-intercept is on the *y*-axis, where *x=0*: *(0,5)*. Plot both and draw the line through them. *(0,−3)* and *(5,0)* put each 0 in the wrong place; *(−3,5)* is a single point, which is not enough for a line and is not on this one; and *(3,0)* drops the minus sign. ***(−3,0)* and *(0,5)*.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 2: Point and Intercept

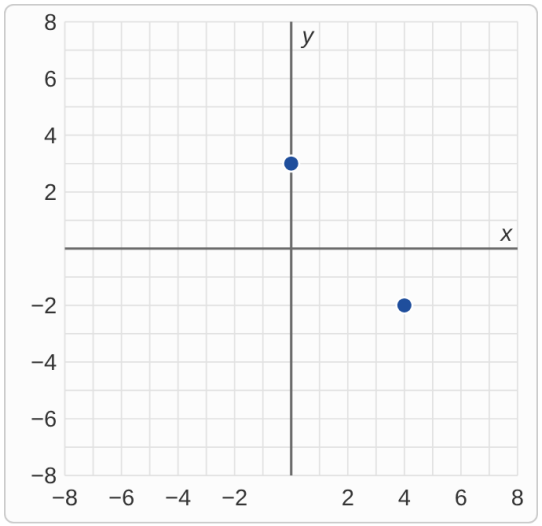


Figure: A grid from −8 to 8 in both directions, with grid lines every 1 unit. Two points are plotted: (0, 3) on the y-axis and (4, −2). No line is drawn yet.

$$(4,-2), \quad y\text{-intercept } 3$$

Question

What is the slope of the line through these two points?

- **A.** $-5/4$
- **B.** $5/4$
- **C.** $-4/5$
- **D.** $4/5$

Answer and reasoning

The y-intercept 3 is the point $(0,3)$. From $(0,3)$ to $(4,-2)$: over 4, down 5, so $m=(-2-3)/(4-0)=-5/4$. You do not need the slope to draw this line, since two points are enough, but it tells you which way the line goes. $5/4$ misses that the line goes down; $-4/5$ is run over rise; and $4/5$ makes both slips: run over rise, and up instead of down. **$-5/4$.**

Example 3: Slope and Intercept

$$m=-2, \quad x\text{-intercept } 3$$

Question

You plot (3, 0). Which point should you plot next?

- **A.** (5, 1)
- **B.** (5, −1)
- **C.** (3, −2)
- **D.** (4, −2)

Answer and reasoning

The *x*-intercept 3 is the point *(3,0)*. A slope of *−2* is *(−2)/1*: right 1, down 2, to *(4,−2)*. Going left 1 and up 2, to *(2,2)*, works too. *(5,1)* moves right 2 and up 1, which swaps rise and run and loses the minus; *(5,−1)* moves right 2 and down 1, which is slope *−1/2*; and *(3,−2)* moves down without moving across, which does not use a run at all. ***(4,−2)*.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 4: Point and Slope

$$m=3, \quad (-2,1)$$

Question

You plot (−2, 1). Which point is also on the line?

- **A.** (−3, 4)
- **B.** (−1, 4)
- **C.** (1, 2)
- **D.** (−2, 4)

Answer and reasoning

Slope 3 means *3/1*: right 1, up 3. From *(−2,1)* that lands on *(−1,4)*. *(−3,4)* goes left 1 and up 3, which is slope *−3*; *(1,2)* goes right 3 and up 1, slope *1/3*; and *(−2,4)* goes straight up, with no run. ***(−1,4).***

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 5: Slope-Intercept Form

$$y=-3x+2$$

Question

How do you graph this line?

A. Start at (2, 0); go right 1, down 3.
B. Start at (0, 2); go right 3, down 1.
C. Start at (0, 2); go right 1, down 3.
D. Start at (2, 0); go right 1, up 3.

Answer and reasoning

In $y=mx+b$, $b=2$ is the y-intercept, the point $(0,2)$. The slope -3 is $(-3)/1$: down 3 for every 1 across. So from $(0,2)$ go right 1 and down 3, to $(1,-1)$. Starting at $(2,0)$ puts the intercept on the wrong axis; right 3, down 1 flips rise and run, which is slope $-1/3$; and $(2,0)$ with up 3 makes two slips, the wrong axis and a lost minus sign. **Start at $(0,2)$; go right 1, down 3.**

Example 6: Point-Slope Form

$$y-2=-3(x+1)$$

Question

Which point and slope does this equation give?

A. The point (−1, 2), slope −3
B. The point (1, −2), slope −3
C. The point (−1, 2), slope 3
D. The point (1, −2), slope 3

Answer and reasoning

Point-slope form is $y-y_1=m(x-x_1)$. Here $y-2$ means $y_1=2$, and $x+1=x-(-1)$ means $x_1=-1$. The slope is the number in front, -3. Plot $(-1,2)$, then go right 1 and down 3. $(1,-2)$ reads the signs as written instead of flipping them; slope 3 drops the minus; and $(1,-2)$ with slope 3 makes both slips. ***(−1,2)*, slope *−3*.**

Example 7: Standard Form 1

$$2x-3y=6$$

Question

Which two intercepts can you plot to graph this line?

A. (3, 0) and (0, 2)
B. (3, 0) and (0, −2)
C. (−3, 0) and (0, 2)
D. (2, 0) and (0, −3)

Answer and reasoning

Set $y=0$: $2x=6$, so $x=3$, the point $(3,0)$. Set $x=0$: $-3y=6$, so $y=-2$, the point $(0,-2)$. Standard form makes intercepts quick. $(0,2)$ loses the sign from dividing by -3; $(-3,0)$ and $(0,2)$ come from changing the sign of the 6, as $2x=-6$ and $-3y=-6$; and $(2,0)$ and $(0,-3)$ divide 6 by the other variable's number. ***(3,0)* and *(0,−2)*.**

Example 8: Standard Form 2

$$3x+4y=4$$

Question

The x-intercept here is 4/3, which is hard to plot. Which two points work better?

A. (0, 1) and (4, −2)
B. (0, 1) and (4, 4)
C. (0, 4) and (4, 1)
D. (0, 1) and (3, −3)

Answer and reasoning

Solve for y: $4y=-3x+4$, so $y=-(3/4)x+1$. Start at the y-intercept $(0,1)$ and use the slope: right 4, down 3, to $(4,-2)$. Check: $3(4)+4(-2)=4$. This is the move to make whenever a standard-form intercept is a fraction. $(4,4)$ goes up instead of down; $(0,4)$ forgets to divide by 4, then moves right 4, down 3 from there; and $(3,-3)$ goes right 3 and down 4, which flips rise and run. ***(0,1)* and *(4,−2)*.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 9: x = c

$$x=-4$$

Question

What does the graph of this line look like?

A. A vertical line through (−4, 0)
B. A horizontal line through (0, −4)
C. A line through the origin with slope −4
D. A horizontal line through (0, 4)

Answer and reasoning

The equation says every point has $x=-4$, whatever y is: $(-4,0)$, $(-4,3)$, $(-4,-7)$. Those points stack straight up and down, so the graph is the vertical line through $(-4,0)$. A horizontal line through $(0,-4)$ is $y=-4$; slope -4 through the origin is $y=-4x$; and a horizontal line through $(0,4)$ is $y=4$, which uses the wrong direction and drops the sign. **A vertical line through $(-4,0)$.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 10: Name the Line

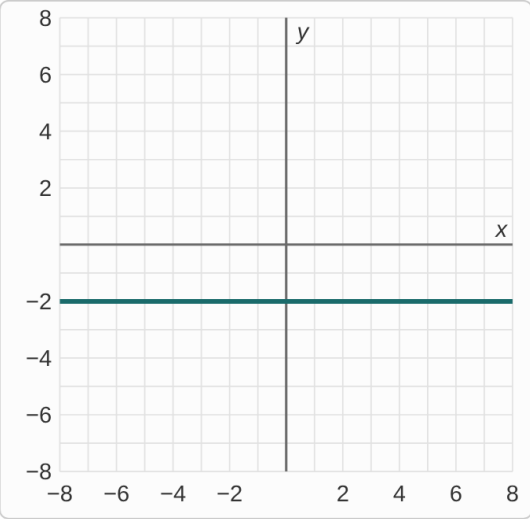


Figure: A straight line on a grid from −8 to 8 in both directions, with grid lines every 1 unit. It runs straight across at height −2, through (−8, −2), (0, −2) and (8, −2).

Question

Which equation does this graph show?

- **A.** $x=-2$
- **B.** $y=-2x$
- **C.** $y=-2$
- **D.** $y=x-2$

Answer and reasoning

Every point on this line has height -2, whatever its x: $(-8,-2)$, $(0,-2)$, $(8,-2)$. That is the equation $y=-2$. $x=-2$ would be an up-and-down line through $(-2,0)$; $y=-2x$ goes through the origin and falls steeply; and $y=x-2$ rises with slope 1. ***y=−2*.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 11: Slope 0

$$m=0, \quad (2,-5)$$

Question

Which is the equation of this line?

A. $x=2$
B. $y=-5$
C. $y=2$
D. $x=-5$

Answer and reasoning

Slope 0 means the height never changes. The line goes through $(2,-5)$, so its height is always -5: $y=-5$. Point-slope form agrees: $y+5=0(x-2)$, so $y=-5$. $x=2$ is the up-and-down line, whose slope is undefined, not 0; $y=2$ uses the wrong coordinate; and $x=-5$ uses the right number but makes an up-and-down line. ***y=−5*.**

Example 12: The Value 0

$$x=0$$

Question

CheckIt can ask you to graph this line. What is its graph?

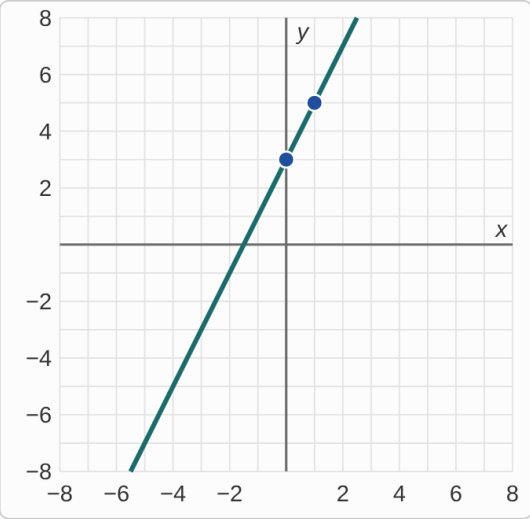
- **A.** The x-axis
- **B.** The y-axis
- **C.** Only the point (0, 0)
- **D.** Nothing: x = 0 has no line to draw

Answer and reasoning

$x=0$ is a vertical line like any $x=c$: every point has x-coordinate 0, such as $(0,-3)$, $(0,0)$ and $(0,5)$. Those are exactly the points on the y-axis. The x-axis is $y=0$; $(0,0)$ is only one of the points, and y can be anything; and 0 is a value like any other, so the line is there. **The *y*-axis.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 13: Slip 1



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (0, 3) and (1, 5) and drew the straight line through them, which goes up from left to right.

$$y=-2x+3$$

Question

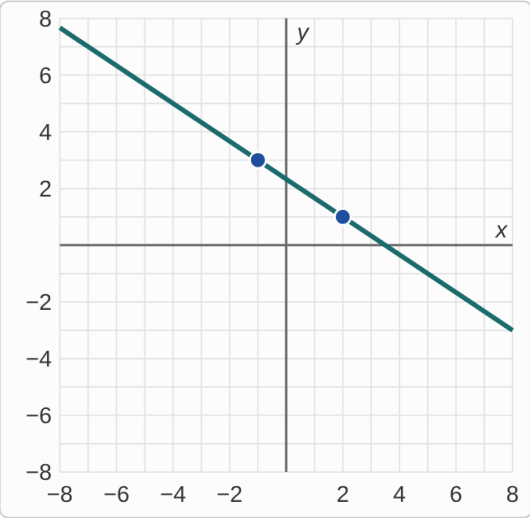
A student graphed this line. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** It should go down: right 1, down 2 from (0, 3).
- **C.** The y-intercept should be the point (3, 0) instead.
- **D.** Slope −2 means right 2, down 1, from (0, 3).

Answer and reasoning

The starting point *(0,3)* is right. But the slope is *−2*, so from *(0,3)* the line goes right 1 and down 2, to *(1,1)*. The student went up, to *(1,5)*, which is slope *+2*. The *y*-intercept is on the *y*-axis, *(0,3)*, not *(3,0)*, which puts it on the wrong axis; and right 2, down 1 flips rise and run, which is slope *−1/2*. **From *(0,3)*, go right 1 and down 2.**

Example 14: Slip 2



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (−1, 3) and (2, 1) and drew the straight line through them.

$$2x+3y=7$$

Question

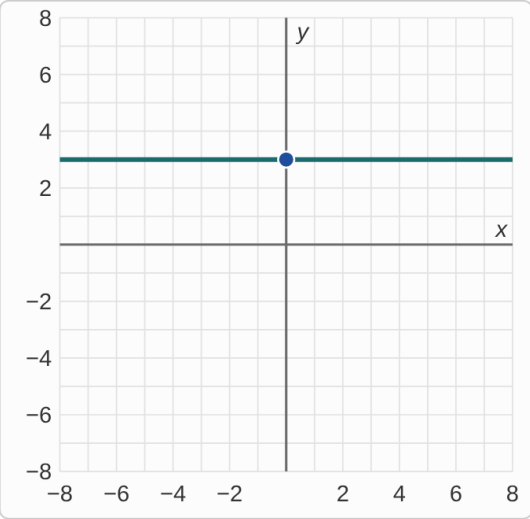
A student graphed this line by finding two points with whole-number coordinates. What is the mistake, if any?

A. There is no mistake.
B. It should use the intercepts, not other points.
C. It should go up, since 2 and 3 are positive.
D. It should cross the y-axis at (0, 7), since C is 7.

Answer and reasoning

Both intercepts are fractions here: $x=0$ gives $y=7/3$, and $y=0$ gives $x=7/2$. So the student tried whole numbers for x. Check both points: $2(-1)+3(3)=7$ and $2(2)+3(1)=7$. The student is right. Any two points on the line work, and these are easier to plot than the intercepts; positive A and B make the slope $-2/3$, so the line goes down; and $(0,7)$ forgets to divide the 7 by 3. **There is no mistake.**

Example 15: Slip 3



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (0, 3) and drew a straight line through it that runs left to right at height 3.

$$x=3$$

Question

A student graphed this line. What is the mistake, if any?

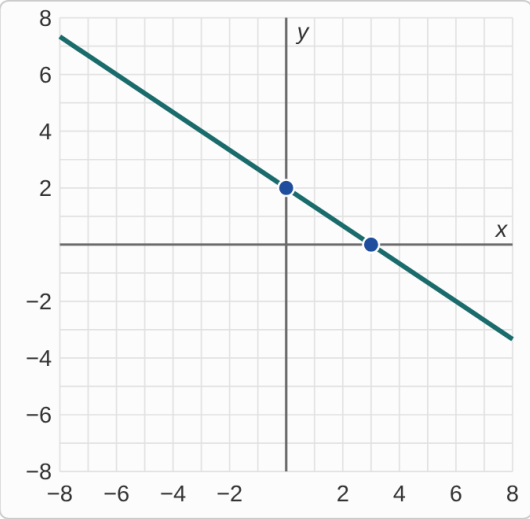
A. There is no mistake.
B. It should be the line through the origin with slope 3.
C. x = 3 is the up-and-down line through (3, 0).
D. It should be the up-and-down line through (0, 3).

Answer and reasoning

$x=3$ says every point has x-coordinate 3: $(3,0)$, $(3,5)$, $(3,-2)$. Those points line up vertically, through $(3,0)$. The student drew $y=3$, the horizontal line. The line through the origin with slope 3 is $y=3x$, which reads the 3 as a slope; and the up-and-down line through $(0,3)$ is the y-axis, $x=0$: it has the right direction but puts the 3 on the wrong axis, as the student did. **Draw the vertical line through $(3,0)$.**

Adapted from TBIL Precalculus (Clontz & Lewis), CC BY-NC-SA 4.0. Drafted with Claude (Anthropic); reviewed by Dr. Marty Romero.

Example 16: Slip 4



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (3, 0) and (0, 2) and drew the straight line through them.

$$3x+2y=6$$

Question

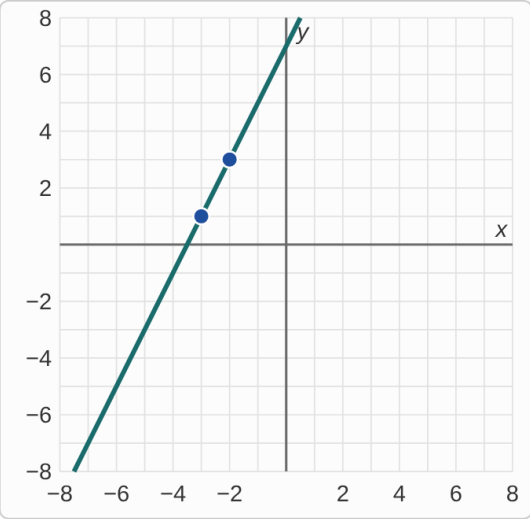
A student graphed this line using its intercepts. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** The line should go up, since both numbers are positive.
- **C.** The intercepts should be (−2, 0) and (0, −3).
- **D.** The intercepts are swapped: they are (2, 0) and (0, 3).

Answer and reasoning

Set $y=0$: $3x=6$, so $x=2$, the point $(2,0)$. Set $x=0$: $2y=6$, so $y=3$, the point $(0,3)$. The student divided 6 by the wrong number each time. Check the student's point: $3(3)+2(0)=9$, not 6. Positive A and B make the line go down, since the slope is $-3/2$; and $(-2,0)$ and $(0,-3)$ would be the intercepts of $3x+2y=-6$, which changes the sign of the 6. **Plot $(2,0)$ and $(0,3)$.**

Example 17: Slip 5



Plotted data: A student's graph on a grid from −8 to 8. The student plotted (−3, 1) and (−2, 3) and drew the straight line through them.

$$y+1=2(x-3)$$

Question

A student graphed this line. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** The point is (3, −1); the student read the signs as written.
- **C.** The slope should be 1/2, so right 2 and up 1.
- **D.** The point should be (−1, 3), with the coordinates swapped.

Answer and reasoning

Point-slope form subtracts the coordinates: $y+1=y-(-1)$, so $y_1=-1$, and $x-3$ means $x_1=3$. The point is $(3,-1)$. The student used $(-3,1)$. The slope 2 was used correctly: right 1, up 2. Slope $1/2$ would flip rise and run; and $(-1,3)$ swaps the coordinates of $(3,-1)$, which gives a point that is not on the line: $3+1=4$, but $2(-1-3)=-8$. **Plot $(3,-1)$, then right 1 and up 2.**